

## Return to Activity and Sport After Injury

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With sports participation comes an inherent risk of injury. In 2017, a summary report on the National High School Sports-Related Injury Surveillance Study for the school year of 2016–17 estimated that 7.9 million students participated in high school sports, with approximately 1.16 million injuries across the United States.<sup>1</sup> Based on this report, the overall injury rate in all high school sports combined was 2.09 injuries per 1000 athlete exposures. Similarly, according to a National Institute of Arthritis and Musculoskeletal and Skin Diseases report, there were more than 2.6 million children younger than 19 years of age who were treated in emergency departments for musculoskeletal injuries due to sport participation.<sup>2,3</sup> In the collegiate setting, Kay et al.<sup>4</sup> reported the number of severe (missing >3 weeks of participation) injuries across 25 National Collegiate Athletic Association (NCAA) sports from 2009 through 2015 to be as high as 3183 injuries over this time, which resulted in an injury rate of 0.66 per 1000 athletic exposures. These statistics help describe the current risk athletes face when participating in sports. While risk exists with sports participation, there are also many physical and psychological benefits that come from sports participation.<sup>5,6</sup> However, previous injuries have been shown over and over again to be a risk factor for future injuries.<sup>7</sup> This creates a challenging situation for sports medicine providers, as safely and successfully returning athletes to sports is generally the long-term goal for patients and providers.

The following chapter is meant to provide a brief overview of the current literature regarding return to sports (RTS) following injury.

Arguably one of the most researched orthopaedic injuries is the anterior cruciate ligament (ACL) injury. In the past 5 years there have been almost 600 review articles published on the topic of ACL injuries, and 50 specifically on the topic of RTS following ACL injury. Despite the plethora of research on the topic, the success of returning to sport following this injury is fair at best. There are variations on what is considered a successful RTS, as the definition of “success” varies among reports. These variations help explain the wide range of data reported in the literature, which currently ranges from 33% to 92% of individuals returning to sport following ACL injury.<sup>8–12</sup> In a systematic review and meta-analysis, Ardern and colleagues<sup>13</sup> reviewed 48 studies with a total subject pool of 5770. They reported that at an average of 3.5 years following ACL reconstruction (ACL-R), 82% of participants returned to some sports participation, 63% had returned to their pre-injury level of participation, and 44%

had returned to competitive sport at an average final follow-up of 41.5 months. One of the contributing factors to the current return-to-sport data is the high rate of re-injury associated with returning to sport following ACL injury. A systematic review by Wiggins et al.<sup>14</sup> reported on the risk of secondary injury following primary ACL injury in young athletes. Nineteen studies were included in the review, with a total number of 72,054 subjects pooled across the studies. Overall, the total second ACL re-injury rate was 15%, with an ipsilateral re-injury rate of 7%, and contralateral injury rate of 8%. The secondary ACL injury rate (ipsilateral and contralateral) for patients younger than 25 years was 21%, and the secondary ACL injury rate for athletes who returned to a sport was 20%. Combining these risk factors, athletes younger than 25 years who returned to sport had a secondary ACL injury rate of 23%.<sup>14</sup> Similarly, Webster et al.<sup>15</sup> examined the second injury risk in those patients less than 20 years of age and found a 29% injury rate (ipsilateral + contralateral) within 5 years following primary ACL reconstruction.

As a result of the high risk of re-injury in the ACL population, there is a need for a comprehensive overview of principles that help guide the clinician’s decision-making process for returning an athlete to sport. For the purposes of this review, the ACL rehabilitation model will be used as an example of specific guidelines and criteria that require consideration when determining an athlete’s readiness for RTS. Once the athlete is able to demonstrate foundational strength of the lower extremity (quadriceps, hamstrings, gluteals, gastrocnemius-soleus complex, etc.), normalized joint ranges of motion at the hip, knee, and ankle, core stability,<sup>16</sup> and the ability to jog/run without noticeable deviations,<sup>17,18</sup> multifactorial functional testing should be administered. Key factors for consideration include a functional testing algorithm (FTA)<sup>17</sup> or test battery<sup>18</sup> involving strength,<sup>19–21</sup> power,<sup>20,21</sup> balance or proprioception,<sup>22,23</sup> movement quality during athletic movements,<sup>24,25</sup> fear of reinjury assessment,<sup>26–28</sup> patient-reported outcomes,<sup>19,29</sup> and limb symmetry indices (LSI).<sup>30</sup>

First, restoration of range of motion (ROM) is imperative. This includes hip, knee, and ankle ROM, as all have been examined in a variety of lower extremity injuries. For example, decreased ankle dorsiflexion (Fig. 35.1) has been associated with increased knee valgus position during landing, which can contribute to a variety of knee injuries.<sup>31,32</sup> In particular, the lack of ankle dorsiflexion may alter lower extremity movement patterns<sup>33</sup> and has been demonstrated to have a relationship with ACL tears.<sup>34</sup> Restoration of knee extension ROM is considered to be important

## Abstract

With sports participation comes an inherent risk of injury. While risk exists with sports participation, there are also many physical and psychological benefits that come from sports participation. Previous injuries have been shown over and over again to be a risk factor for future injuries. This creates a challenging situation for sports medicine providers, as safely and successfully returning athletes to sports is generally the long-term goal for patients and providers. As a result of the high risk of re-injury in the anterior cruciate ligament (ACL) population, there is a need for a comprehensive overview of principles that help guide the clinician's decision-making process for returning an athlete to sport. For the purposes of this review, the ACL rehabilitation model will be used as an example of specific guidelines and criteria that require consideration when determining an athlete's readiness for return to sport. Once the athlete is able to demonstrate foundational strength of the lower extremity (quadriceps, hamstrings, gluteals, gastrocnemius-soleus complex, etc.), normalized joint ranges of motion at the hip, knee and ankle, core stability, and the ability to jog/run without noticeable deviations, multifactorial functional testing should be administered. Key factors for consideration include a functional testing algorithm (FTA) or test battery involving strength, power, balance or proprioception, movement quality during athletic movements, fear of re-injury assessment, patient-reported outcomes, and limb symmetry indices (LSI). In addition, the concept of progressive loading should be given consideration as the athlete is gradually returned to sport.

## Keywords

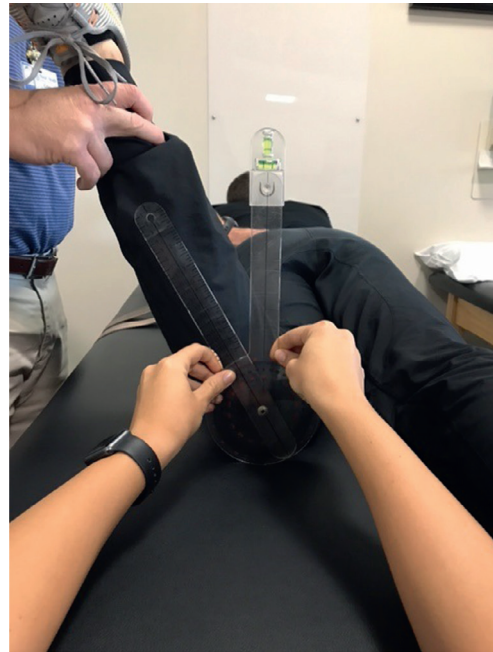
return to sport  
strength  
neuromuscular control  
progressive loading  
movement assessment  
power



**Fig. 35.1** Measurement of ankle dorsiflexion range of motion. The ankle is placed in a subtalar neutral position and the foot is aligned with a line marked on the floor. The line on the floor will be continuous with a line marked on the wall. The participant is instructed to bring the testing knee toward the line on the wall until he or she feels the ankle no longer goes into dorsiflexion without lifting the heel off the ground. The investigator places the top of the long base of the inclinometer on the tibial tuberosity to read the angle from the vertical line. The angle between the tibia and the vertical line is recorded.

following ACL-R in both the early and late stages of the rehabilitation process.<sup>35,36</sup> Decreased knee extension ROM following ACL injuries has been found to be a contributing factor to re-injury and to the development of osteoarthritis at a 20-year follow-up.<sup>37</sup> Decreased hip ROM (Fig. 35.2) has been found to correlate with hip, knee, and low back pain,<sup>38–40</sup> and may be associated with ACL injury.<sup>41–43</sup> In summary, restoration of ROM of the entire lower extremity to be equal to or similar to the uninvolved side should be an expectation prior to return-to-sports clearance.

In addition to ROM, restoration of lower extremity strength is considered a staple in return-to-sports testing. Deficits in gluteals, quadriceps, and hamstrings muscle strength have been shown to be correlated with changes in lower extremity kinematics and in some cases may be attributed as risk factors for both initial injury and re-injury.<sup>44–48</sup> The use of handheld dynamometry (Fig. 35.3) and isokinetic testing (Fig. 35.4) provides a more accurate means of muscle strength testing and should be encouraged over manual muscle testing when available. A 90% quadriceps and hamstrings limb symmetry index ( $LSI = \text{involved limb/uninvolved limb} \times 100$ ) at the time of RTS has been suggested as the minimal acceptable cutoff as a passing score for RTS, while some suggest a symmetry of 100% as an acceptable cutoff.<sup>17,21</sup> In addition to normalizing strength LSI, normalizing agonist and antagonist muscle function around the joint should also be considered. In the context of the ACL, this is typically looked at in the framework of quadriceps to hamstrings ratios. Literature recommends a ratio greater than 55% for females and greater than 62.5% for males as a minimum passing score for



**Fig. 35.2** Measurement of hip range of motion (ROM). Participant lays prone on the table with the knee flexed to 90 degrees while the investigator stabilizes the hip and passively moves the lower extremity through full internal and external ROM stopping just prior to movement at the hip.



**Fig. 35.3** Measurement of hip external rotator strength using a handheld dynamometer. Participant lays prone on the table with the knee flexed to 90 degrees while the investigator stabilizes the hip and places the dynamometer just superior to the medial malleolus. The participant is instructed to rotate the lower leg toward the midline of his or her body with maximal effort while the investigator meets the resistance of the moving lower leg.

RTS.<sup>49,50</sup> While the current review is using the ACL as a framework for this discussion, this concept can be extrapolated to the shoulder, where maintaining adequate shoulder external rotator to internal rotator cuff strength (66% to 75% ratio) may help minimize injury and/or re-injury risk.<sup>51</sup>





**Fig. 35.4** Isokinetic strength testing. The participant is seated on the isokinetic dynamometer and secured with padded straps around the thigh, pelvis, and torso to minimize accessory and compensatory movements during testing. The participant is instructed to either push (knee extension) or pull (knee flexion) against the arm of the dynamometer with maximal effort for a total of five repetitions. Isokinetic testing is often ordered at 60 degrees/s and 180 degrees/s for both the quadriceps and hamstrings. A symmetry index of 90% or greater (involved limb/uninvolved limb) is considered acceptable for return to sport. In addition, the time to peak torque should be examined as this is a measure of how long it takes for each limb to produce the highest peak value during the test. If the peak torque value is similar or the same between limbs, but the involved limb takes twice as long to produce that force, this should be noted as an inability for the participant to produce force quickly, which is needed for dynamic athletic movements.

In conjunction with lower extremity ROM and strength, it is important to assess a patient's ability to produce power. This is especially important in sports that require quick and explosive movements. Assessing power has historically been done through the use of hop testing. Numerous jump tests are utilized throughout the literature including single hop,<sup>52–54</sup> triple hop,<sup>52,54,55</sup> crossover hop,<sup>54,56,57</sup> and timed hop.<sup>54,55,57</sup> Overall, a 90% LSI between limbs is considered to be acceptable and may be used as criteria for RTS.<sup>56,58–60</sup> There are many limitations to these tests, including the lack of assessment for movement quality and potential compensation strategies that patients may utilize as a means to complete the test; however, these tests (Fig. 35.5) are well supported in the literature and should be incorporated as part of a comprehensive evaluation when considering return to sport following ACL-R.<sup>21</sup>

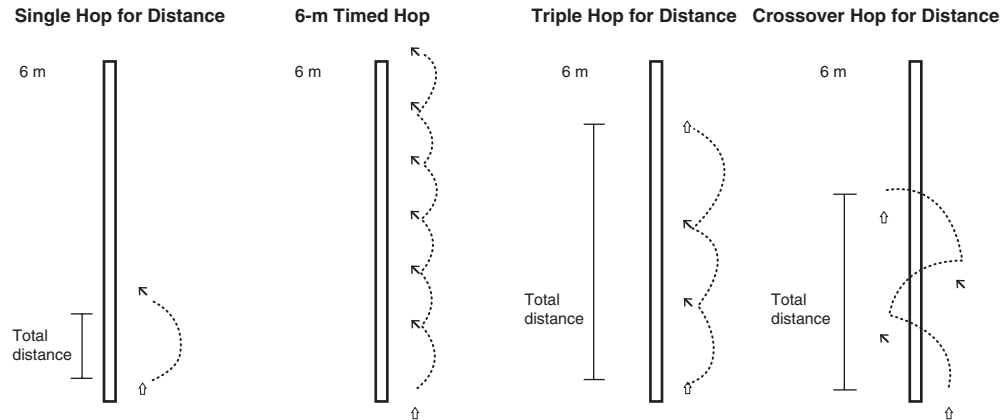
Beyond the aforementioned criteria for RTS, movement quality should be considered, as numerous studies support the notion that poor lower extremity balance and movement patterns are associated with primary and secondary injury.<sup>24,61–65</sup> For example, performance on the Y-Balance Test has been shown to be an accurate screening tool for injury in numerous populations.<sup>65–67</sup> A cutoff of 94% composite score is considered the minimal acceptable threshold for passing this test, as scores below this threshold have been associated with increased injury risk.<sup>65</sup> Furthermore, anterior reach asymmetry on the Y-Balance Test (Fig. 35.6) has been shown to be correlated with performance during return to sport testing,<sup>23</sup> and a side-to-side asymmetry greater than 4 cm is considered a higher risk for lower extremity injury.<sup>67</sup>

In the ACL-R population at time of RTS, participants have previously demonstrated an anterior reach (ANT) of 97% LSI with individual components of 65% of leg length (LL) for ANT, 99% LL for posteromedial (PM), and 94% LL for posterolateral (PL).<sup>22</sup> Beyond the Y-Balance Test, additional screening tools should be applied for assessment of movement quality during dynamic activities. The jump-landing task has been used as a lower extremity screening tool, and performance on this task is associated with primary and secondary injury.<sup>24</sup> The most commonly used assessment of the jump-landing task is through the use of the Landing Error Scoring System (LESS).<sup>24</sup> This test is commonly utilized in the literature as a screening tool for lower extremity injury and is well supported as both an injury risk screen and return-to-sports test.<sup>24,68,69</sup> Deficits of decreased knee flexion, increased knee valgus positioning, and decreased trunk control during this task have been associated with primary and secondary ACL injury.<sup>24,68,69</sup> Overall, demonstration of asymmetry between limbs during dynamic movements could be harmful and should be given consideration.

As sporting movements tend to require the athlete to move in multiple directions, an assessment of movement in multiple planes is warranted. The Vail Sport Test assesses dynamic movements in multiple planes of motion and requires the athlete to move through both the frontal and sagittal planes while continuing to demonstrate vertical excursions.<sup>25</sup> Four components comprise the Vail Sport Test, and include single leg squat for 3 minutes, lateral bounding for 90 seconds, and forward/backward jogging for 2 minutes each. After each component, the participant is given 2.5 minutes of rest prior to proceeding to the next task.<sup>25</sup> Finally, the Vail Sport Test also incorporates external perturbation to the athlete during the testing procedure through the use of sport cord resistance. This test has been shown to be reliable and is recommended as part of a return-to-sport test battery with a passing score of at least 46/54.<sup>25</sup> Likewise, if a clinician has the capabilities for assessment of dynamic movement using 3D motion capture (Fig. 35.7) at time of return to sport, this information may provide in-depth analysis of an athlete's potential movement strategies that may be compensatory in nature and contribute to future injury.<sup>64,70,71</sup> Previous research has demonstrated that deficits in core stability,<sup>72</sup> high knee abduction angles and moments,<sup>64,73</sup> and low knee flexion angles<sup>73</sup> may place one at a greater risk for ACL injury. The use of 3D motion capture has previously been used at 6 months postoperative ACL-R and time of RTS, and findings of decreased knee flexion angles during gait,<sup>74</sup> running,<sup>75</sup> and landing<sup>76</sup> have been demonstrated. Furthermore, ACL-R individuals who were tested at time of RTS and demonstrated increased knee valgus, deficits in hip external rotation muscle torque, and asymmetries in knee moments during landing activities were at a significantly higher risk for a second ACL injury.<sup>64</sup> As such, the use of 3D motion capture can be a useful tool in determining an athlete's readiness for return to sport following ACL-R.

In addition to the previously mentioned objective measurements of physical performance, assessment of the athlete on a cognitive level is also warranted. Successfully returning to sport following ACL injury at 12 months postsurgery has been linked to a positive psychological state.<sup>26</sup> Prior research has established that psychological readiness to return to sport and recreation

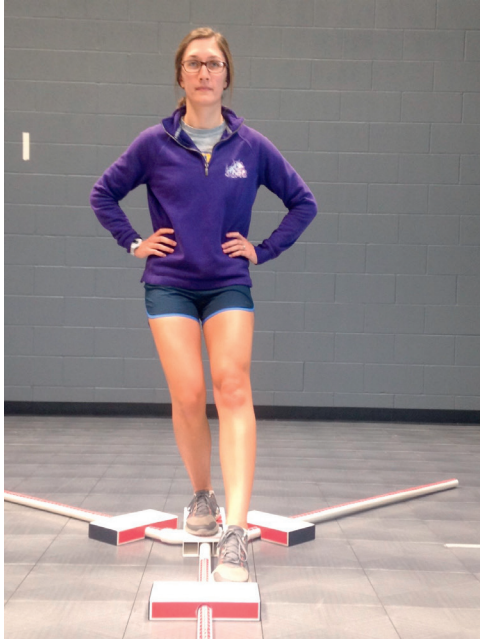
Diagrammatic representation of the series of 4 hop tests: single hop for distance, 6-m timed hop, triple hop for distance, and crossover hop for distance.



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**Fig. 35.5** Hop tests. *Single hop for distance*—The participant is instructed to stand on the limb to be tested and to hop off that limb with maximal effort while landing on the same limb. *6-m timed hop*—The participant is instructed to stand on the limb to be tested and to hop off that limb while performing repeated hops for the total distance as quickly as possible. *Triple hop for distance*—The participant is instructed to stand on the limb to be tested and to hop off that limb with maximal effort for three consecutive hops. *Crossover hop for distance*—The participant is instructed to stand on the limb to be tested and to hop off that limb with maximal effort for three consecutive hops while alternating crossing over a mark on the floor. For each hop test, the total distance covered during the hopping motion is recorded while the time taken to cover the 6 m is recorded for the timed hop. (Reid A, Birmingham TB, Stratford PW, Alcock GK, Giffin JR. Hop testing provides a reliable and valid outcome measure during rehabilitation after anterior cruciate ligament reconstruction. *Phys Ther.* 2007;87[3]:337–349.)



**Fig. 35.6** Y-balance test anterior reach. The participant is instructed to perform a single limb stance on the extremity being tested while reaching outside their base of support with the uninvolvement limb to push a reach indicator box along the measurement pipe.



**Fig. 35.7** 3D motion capture of dynamic movement. Dynamic movements that replicate athletic movements seen on the field or court of play can be tested in the motion capture lab. Tasks such as box drop jump landings, forward hops, lateral hops, and cutting maneuvers can all be tested to determine a participant's quality of movement. Key indicators of faulty movement patterns include increased knee valgus angles and joint moments, decreased knee flexion angles, and overall asymmetry between limbs during the landing phase of the movement.

was the factor most strongly associated with returning to the preinjury activity.<sup>77</sup> One scale that is used to measure an athlete's psychological state at time of RTS following ACL-R is the ACL-Return to Sport After Injury (ACL-RSI) scale.<sup>28,60</sup> The ACL-RSI has been shown to be reliable and valid in a population of athletes considering a return to sport following ACL-R. The scale may be effective in discriminating between general confidence in returning to the sport and confidence in the injured knee with returning to sport.<sup>28</sup> Similarly, the ACL-RSI has demonstrated utility in predicting capabilities of returning to sport following ACL-R from a psychological perspective with those scoring greater than 76 returning at a higher rate.<sup>78</sup> These studies indicate that patient perception and patient self-rating may play a large role in achieving successful outcomes. The International Knee Documentation Committee short form (IKDC) is used throughout the literature as a patient reported outcome following injury.<sup>60</sup> The possible score of the IKDC ranges from 0 to 100, with a higher score indicating a higher level of functioning. The IKDC has been shown to be correlated with functional outcome measures such as quadriceps strength,<sup>19,79,80</sup> hop tests,<sup>81,82</sup> and has also been shown to have good psychometric properties, making it an ideal patient-reported outcome.<sup>83–85</sup> Furthermore, the IKDC has been shown to discriminate between successful and unsuccessful performance on return-to-sport testing, with scores greater than 90 having more success.<sup>29,86</sup>

The final consideration that needs to be made when faced with the challenge of returning an athlete to sport is the principle of progressive loading. This basic principle is founded in physical training and at its simplest requires a gradual increase in volume, intensity, frequency, or time in order to achieve the targeted goal.<sup>87</sup> In the context of returning to sport, these principles must be balanced across rehabilitation, strength and conditioning,

practices and games to maintain a progressive overall load until a successful return has been achieved.<sup>88,89</sup> The concept of acute load (all physical activity done in a 1 week time frame) should be balanced with the chronic load (all physical activity done over the past 4 weeks). When spikes in acute load occur above what the chronic load has been, the risk of injury significantly increases (Fig. 35.8).<sup>88</sup> This concept of load needs to be considered by all members of the sports medicine team, and proper communication between disciplines is paramount to correctly managing load as an athlete prepares for return to sport. In general, an acute to chronic training load that falls between 0.8 and 1.3 (i.e., loads are approximately equal) carries a relatively low risk of injury.<sup>88,90</sup> When the training load exceeds 1.5, then athletes may be at a greater risk of suffering injury (Fig. 35.9).

## CONCLUSION

The previous discussion outlines a thorough overview of components that need to be considered from a physical and psychological standpoint when considering return to activity following injury (Table 35.1). While this discussion focused on ACL injury, the principles of assessing ROM, strength, power, movement quality, and psychological readiness can and should be extrapolated to any athlete returning to activity/sport following injury.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

## SELECTED READINGS

### Citation:

Davies GJ, McCarty E, Provencher M, et al. ACL return to sport guidelines and criteria. *Curr Rev Musculoskelet Med*. 2017;10(7):307–314.

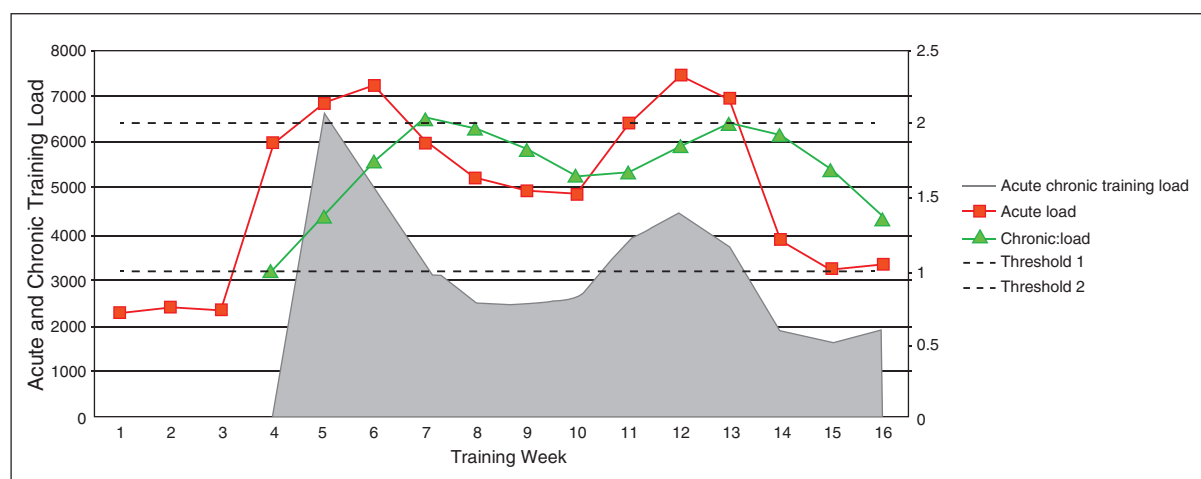
**TABLE 35.1 Return to Sport Considerations Following Anterior Cruciate Ligament Reconstruction**

|                     |                                                                                                                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROM                 | • ROM of hip, knee, and ankle equal to or similar to the uninvolved side. Goal of side to side symmetry                                                                                                        |
| • Hip               | • Decreased hip IR associated with ACL injury <sup>41–43</sup>                                                                                                                                                 |
| • Knee              | • ≤5 degrees side-to-side difference in knee extension <sup>35,36</sup>                                                                                                                                        |
| • Ankle             | • Ankle DF deficits alter lower extremity movement patterns and are associated with ACL injury <sup>33,34</sup>                                                                                                |
| Strength            | • 90% or greater LSI of quadriceps and hamstrings at time of RTS <sup>17,21</sup>                                                                                                                              |
| • Quadriceps        | • Quadriceps to Hamstrings ratio at time of RTS: >55% for females, and >62.5% for males <sup>49,50</sup>                                                                                                       |
| • Hamstrings        | • 90% or greater LSI of gluteals at time of RTS                                                                                                                                                                |
| • Gluteals          |                                                                                                                                                                                                                |
| Power               | • 90% or greater LSI at time of RTS <sup>56,58,59</sup>                                                                                                                                                        |
| • Hop Tests         |                                                                                                                                                                                                                |
| Movement Quality    | • <94% Y-Balance Test composite score related to increase risk of lower extremity injury <sup>64</sup>                                                                                                         |
| • Y-Balance Test    | • Y-Balance Test side to side anterior reach >4 cm related to increase risk of lower extremity injury <sup>64,66</sup>                                                                                         |
| • LESS              | • Y-Balance Test anterior reach LSI of 97% with 65% of LL for anterior, 99% of LL for posteromedial, and 94% of LL for posterolateral <sup>22</sup>                                                            |
| • Vail Sport Test   | • LESS score >6 = poor; LESS score ≤4 = excellent <sup>24</sup>                                                                                                                                                |
| • 3D Motion Capture | • 46/54 passing score for RTS <sup>25</sup>                                                                                                                                                                    |
|                     | • Assessment of increased knee valgus angles and moments, <sup>63,72</sup> and decreased knee flexion angles <sup>72–75</sup> during dynamic activities that may lead to increased risk of injury or re-injury |

This table provides objective measurements for the clinician to consider prior to making a decision about the participant's ability to return to sport. These factors should be considered within the context of other variables such as patient-reported outcomes (IKDC, ACL-RSI) and acute to chronic workload ratio.

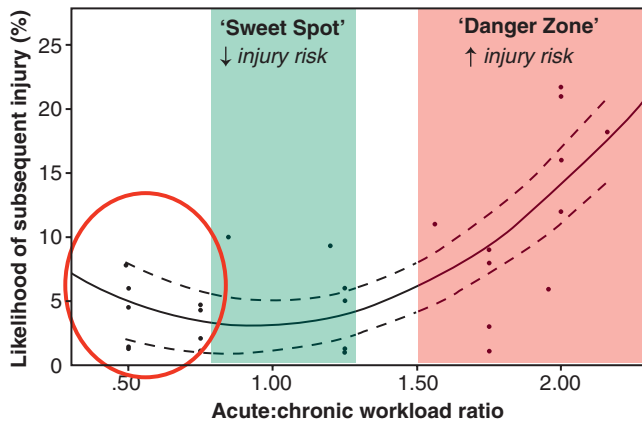
ACL, Anterior cruciate ligament; IR, internal rotation; LL, leg length; LESS, landing error scoring system; LSI, limb symmetry indices; ROM, range of motion; RSI, Return to Sport After Injury.

| Name          | Athlete A                    |                                  |                                                                               |
|---------------|------------------------------|----------------------------------|-------------------------------------------------------------------------------|
| Training Week | Acute Load<br>(Current week) | Chronic Load<br>(4 week average) | Acute:Chronic Training Load Ratio<br>(Previous 4 week average / current week) |
| 1             | 2300                         |                                  |                                                                               |
| 2             | 2420                         |                                  |                                                                               |
| 3             | 2357                         |                                  |                                                                               |
| 4             | 6000                         | 3269.3                           |                                                                               |
| 5             | 6856                         | 4408.3                           | 2.10                                                                          |
| 6             | 7236                         | 5612.3                           | 1.64                                                                          |
| 7             | 6000                         | 6523.0                           | 1.07                                                                          |
| 8             | 5199                         | 6322.8                           | 0.80                                                                          |
| 9             | 4919                         | 6838.5                           | 0.78                                                                          |
| 10            | 4816                         | 5244.8                           | 0.83                                                                          |
| 11            | 6432                         | 5352.8                           | 1.23                                                                          |
| 12            | 7450                         | 5915.5                           | 1.39                                                                          |
| 13            | 6942                         | 6421.3                           | 1.17                                                                          |
| 14            | 3856                         | 6170.0                           | 0.60                                                                          |
| 15            | 3237                         | 5371.3                           | 0.52                                                                          |
| 16            | 3324                         | 4339.8                           | 0.62                                                                          |



**Fig. 35.8** Acute versus chronic training load. This table represents an example of a 16-week training load schedule for both the Acute (current week) and Chronic (4 weeks average) phases. The training load is calculated by multiplying the number of minutes in which participation in the activity occurred (internal load) and/or the external load (i.e., distance covered, balls thrown or hit, jumps, etc.) by the rating of perceived exertion (RPE) of the participant during the activity. This calculation provides the weekly value for total training load. The acute to chronic workload ratio can be calculated by dividing the current week's training load by the previous 4 weeks' training load average (acute load/chronic load).





**Fig. 35.9** Acute to chronic workload ratio. In general, this ratio should be close to 1.0. An acute to chronic training load that falls between 0.8 and 1.3 (i.e., loads are approximately equal) carries a relatively low risk of injury, while a training load that exceeds 1.5 may represent a greater risk of suffering injury.

#### Level of Evidence:

V

#### Summary:

This article provides functional clinical insight into evidence-based criteria that can be used to identify when an individual is ready to return to sport following anterior cruciate ligament reconstruction. An overview of functional tests is framed within an algorithm that helps clinicians qualitatively and quantitatively make clinical decisions.

#### Citation:

Arden CL, Webster KE, Taylor NF, et al. Return to sport following anterior cruciate ligament reconstruction surgery: a systematic

review and meta-analysis of the state of play. *Br J Sports Med.* 2011;45(7):596–606.

#### Level of Evidence:

I

#### Summary:

This meta-analysis and systematic review describes key factors that determine readiness for return to sport following anterior cruciate ligament reconstruction. These findings are based on objective data from more than 5500 participants who were attempting a return to sport. In addition, the results suggest that return-to-sport levels are not as successful as one might believe, and one of the factors that is lacking might involve a psychological component.

#### Citation:

Welling W, Benjaminse A, Seil R, et al. Low rates of patients meeting return to sport criteria 9 months after anterior cruciate ligament reconstruction: a prospective longitudinal study. *Knee Surg Sports Traumatol Arthrosc.* 2018. doi:10.1007/s00167-018-4916-4.

#### Level of Evidence:

III

#### Summary:

An overview of objective and subjective changes that occur in individuals from 6 to 9 months following anterior cruciate ligament reconstruction suggests that time may play an important factor in successful return to sport. This article highlights the lack of readiness of return to sport in patients who have undergone anterior cruciate ligament reconstruction secondary to deficits in quadriceps strength and patient-reported perception of function.



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